

Defense Q&A Preparation

The ten hardest expected questions + model answers

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Q1. Why soft voting instead of stacking or learned weights?

Question

“You chose soft voting to average four model probabilities. Why not learn ensemble weights (stacking, a meta-learner)? Would that not be better?”

Answer

I considered all three options and chose soft voting for **three practical reasons**:

First, stacking needs a second validation set to fit the meta-learner. If I reuse the same val set already used for checkpoint selection, the meta-learner overfits the *quirks* of that val set rather than learning real signal. The ISIC 2018 split is already small; taking a slice for stacking means removing it from train, which is not worth the trade-off.

Second, hard voting (each model takes argmax then votes) discards probability information. Soft voting keeps the full vector $p_m(x) \in [0, 1]^7$, which lets the ensemble distinguish “all four models are very sure of MEL” from “two models vote MEL 51%, two vote NV 49%”. This distinction matters clinically.

Third, soft voting already reaches 89.49% in-distribution and 90.49% with metadata. McNemar’s test shows soft voting beats every single model with $p < 10^{-7}$ against the strongest backbone. There is no clear *headroom* for stacking to break through, and stacking adds inference cost.

Delivery tip

If the committee pushes back with “I still think stacking would be better” — concede that it might, but say “I prioritised reproducibility and simplicity at a result that already matches SOTA. Stacking is the next direction if more time and data become available.”

Q2. How do you know your Grad-CAM is “correct”?

Question

“A red-and-yellow heatmap looks pretty to anyone. How do you know the model is genuinely looking at the lesion and not at some artefact that happens to overlap?”

Answer

I do not stop at qualitative inspection. I measure **four quantitative metrics** against a lesion mask (generated by Otsu thresholding) on the *entire test set*:

- **focus_ratio** — fraction of total activation inside the lesion mask. High = correct focus. All four models reach > 0.6 .
- **peak_distance** — distance from peak activation to lesion centroid (normalised by image diagonal). Low = focus on the centre. All four models reach < 0.2 .
- **entropy** — Shannon entropy of the heatmap. Low = sharp focus.
- **coverage_75** — % area containing the top-25% activation. Low = compact.

This is an improvement over many works that only show qualitative heatmaps without measurement. I learned the importance of quantification the hard way: my first attempt used `model.bn2` for EfficientNet and **focus_ratio** was only 0.18 — the heatmaps were uniformly blue. I switched to `blocks[-1][-1]` and **focus_ratio** jumped to 0.51 — only then did the heatmaps actually outline the lesion. Without the metric, the `bn2` mistake would have slipped past visual inspection.

Delivery tip

Emphasise the “quantitative measurement caught a bug” story — it shows scientific thinking, not just running code.

Q3. How do Macro F1 and balanced accuracy differ? Why choose Macro F1?

Question

“You report Macro F1 rather than balanced accuracy. How do they differ? Why this choice?”

Answer

Balanced accuracy = mean of per-class recall. It looks only at recall and ignores precision.

Macro F1 = mean of per-class F1 = mean of the harmonic mean of *both* precision and recall.

Practical difference: balanced accuracy only penalises *misses* (false negatives). Macro F1 also penalises *false alarms* (false positives). On data with 58:1 imbalance, a model can achieve high balanced accuracy by predicting DF for every ambiguous image — DF recall reaches 100%, balanced accuracy rises, but DF precision collapses. Macro F1 catches that.

I chose Macro F1 because both error types have clinical cost: missing MEL is dangerous, but false alarms also trigger unnecessary biopsies, raising costs and stressing patients.

I still report Sensitivity (= Recall) and Specificity (= $1 - \text{false positive rate}$) per class for the complete picture.

Q4. Why does Swin-Tiny overfit faster than the CNNs?

Question

“Swin-Tiny early-stops at epoch 13 while the CNNs train to epochs 39–49. Why?”

Answer

Three cumulative reasons:

(1) Lack of inductive bias. CNNs come with translation invariance and locality built in — convolution kernels enforce them by construction. A Transformer learns both from data, so it needs more data to converge. ISIC 2018 has only 7,010 training images — small by computer-vision standards (ImageNet has 1.28M).

(2) Extremely high parameter-to-sample ratio. Swin-Tiny has 28M parameters and the training set has 7,010 images, giving a ratio of $4,000\times$. That is extremely overfit-prone. EB0's 5.3M parameters give $756\times$, much more reasonable.

(3) Quadratic self-attention. Every pair of patches attends to every other pair, making it easy to memorise specific spatial structures of the training images.

I mitigated as much as I could: learning rate $1e-4$ (CNNs use $3e-4$) to preserve the pretrained attention weights, weight decay $5e-2$ (CNNs use $1e-3$) for stronger regularisation, and a 5-epoch warmup (CNNs use 3). Early stopping with patience 12 catches the epoch-13 checkpoint.

The silver lining: Swin's *multi-seed mean* accuracy is the highest of any single model (88.33% vs. EB0 87.31%). Early stopping does not mean low performance — it means *higher variance*. In the ensemble, high variance combined with uncorrelated errors against the CNNs makes Swin a valuable ensemble component.

Q5. What happens on images outside the 7 classes (e.g. acne)?

Question

“You only trained on 7 classes. If a patient photographs acne, a scar, or some non-lesion — what does the model do?”

Answer

The model is *forced* to assign one of the 7 classes because the softmax outputs sum to 1 — there is no “other” class. This is the genuine limitation of any closed-set classifier.

I integrated two mitigations:

Confidence threshold. The Streamlit app uses a 50% threshold; if the top-1 probability is below 50%, it shows a “Low confidence” warning and suggests consulting a specialist. In the ISIC test set the low-confidence rate is only $\sim 2\%$, so the warning does not spam the user.

Predictive entropy. The app also computes the entropy of the probability vector. High entropy (near $\log 7$) means the model is unsure between many classes — a possible OOD signal. The app displays entropy numerically.

A more complete production solution would be a dedicated *open-set classifier* or *OOD detector* (e.g. Mahalanobis distance on features, or an Energy-based score). I have not implemented this — it is the obvious next step.

Delivery tip

This is a question a clinical or AI-focused examiner will care about. Acknowledge the limitation honestly and explain mitigation. Saying “the model will classify into the closest class” is wrong.

Q6. How many false-negative MEL cases are there? Clinical impact?

Question

“You report a melanoma F1 of 0.744. Concretely, out of 167 MEL test images, how many does the model miss? What is the clinical impact?”

Answer

Numbers on the ISIC 2018 test set:

- Test set contains 167 actual MEL images.
- Ensemble correctly identifies 124 → **43 false negatives** (recall 0.740).
- Among the 43 misses: 35 are misclassified as NV (benign nevus), 8 as BKL.

Clinical impact if deployed as-is: for every 100 actual MEL patients, 74 would be correctly flagged, 26 would be labelled “benign” and risk leaving the clinic without a biopsy. This is exactly why the Streamlit prototype **never** replaces a clinician — it is a second opinion.

Mitigation is available: threshold tuning with Youden’s J (slide A18) lowers the MEL decision threshold from 0.143 (default) to 0.100, raising MEL Sensitivity from 0.74 to 0.83 — misses drop from 26/100 to 17/100. The trade-off is more false positives (specificity drops from 0.97 to 0.95 — acceptable clinically, since missing MEL is much more dangerous than a false alarm).

Q7. Why not Focal Loss instead of Label Smoothing?

Question

“Focal Loss is designed for imbalanced data. Did you try it? Why pick Label Smoothing?”

Answer

I did try both. The `src/models/losses.py` file contains a `FocalLoss` class — ready to use. My reasons for choosing Label Smoothing in the final pipeline:

Focal Loss ($FL(p) = -(1 - p)^\gamma \log p$) increases the weight on *hard samples* within a batch. In a setup where I already have Weighted Sampler balancing classes plus Mixup/CutMix producing soft mixed labels, Focal Loss creates *complex interactions*: a Mixup-mixed sample has a soft label like $[0, \dots, 0.7, \dots, 0.3]$, and Focal Loss does not have a canonical formula for soft labels — I would have to hand-craft one. Result: F1 fluctuates and is less stable than cross-entropy plus label smoothing.

Label Smoothing is simpler: it multiplies the one-hot label by $(1 - \epsilon)$ and spreads ϵ/C across all classes. PyTorch’s standard cross-entropy supports soft labels natively (`label_smoothing` argument), so it composes cleanly with Mixup/CutMix. The ablation shows removing Label Smoothing drops F1 by 3.11%.

In short: the “Weighted Sampler + Mixup/CutMix + Label Smoothing” recipe is enough to handle imbalance (final 89.5% Acc, 0.845 F1) without needing Focal Loss. If forced to choose just one of the two *without* Mixup, Focal Loss might win.

Q8. What is your ECE? Is the model calibrated?

Question

“You report a high Macro AUC of 0.98. But AUC only measures ranking, not calibration. What is your ECE?”

Answer

I measured Expected Calibration Error (ECE) for all 4 models and applied temperature scaling (Guo et al., 2017):

Model	ECE before	ECE after temperature scaling
EfficientNet-B0	0.041	0.018
ResNet50	0.052	0.022
DenseNet121	0.039	0.019
Swin-Tiny	0.067	0.024

Before calibration: ECE 4–7% — the models are mildly overconfident, especially Swin (6.7%). When the model says “MEL 90%”, the actual accuracy in that confidence bucket is only ~83%.

After temperature scaling: ECE drops to 2–2.4% — the probabilities are now trustworthy enough for clinical decisions. Ensembling calibrated models lowers ECE further.

Code: `src/utils/calibration.py` and `src/calibration_analysis.py`.

Q9. The dataset is mostly fair-skinned. Demographic bias impact?

Question

“HAM10000 was collected in Austria and the USA — predominantly fair skin (Fitzpatrick I–III). If you deploy in Vietnam (Fitzpatrick III–V), what happens?”

Answer

This is a genuine limitation of my thesis, which I acknowledge and discuss explicitly. Three points:

(1) Skin-tone bias is a known field-wide issue. Adamson & Smith (2018) showed that commercial dermoscopy AI (such as SkinVision) under-performs significantly on darker skin — MEL recall drops by ~20% on Fitzpatrick V–VI versus I–II. The field does not yet have a comprehensive solution.

(2) I do not yet have Vietnamese data to test directly. But the OOD evaluation on PAD-UFES-20 (Brazil, with more Fitzpatrick III–IV than HAM10000) shows the model **drops 56 pp in accuracy**. A meaningful portion of that drop *may* come from skin-tone shift; another portion comes from modality shift (smartphone vs dermoscope). I cannot disentangle the two contributions.

(3) Plausible remediation paths:

- Fine-tune on Vietnamese data once available (~500 images would suffice).
- Skin-tone-aware colour augmentation: shift hue/saturation according to the Fitzpatrick scale.
- Group fairness loss (reweight the loss by skin-tone subgroup).

I do not claim the model is ready for Vietnam without adaptation. The Streamlit prototype is a *research prototype* with a clear disclaimer.

Delivery tip

This is an **ethics question**. Answer by *accepting responsibility*, not by arguing or deflecting blame.

Q10. What more does production deployment require?

Question

“If a hospital wanted to actually deploy this system, what else is needed beyond what you have built?”

Answer

I have thought through the steps:

1. Regulatory compliance. Medical diagnostic software needs CE marking (EU MDR) or FDA 510(k) clearance — the prototype does not. Equivalent systems like SkinVision and MoleScope went through this process over 1–2 years.

2. In-domain data. As discussed in Q9, ~500–5000 images from the actual hospital, taken by its own staff, are needed for fine-tuning to reduce domain shift.

3. Open-set / OOD detection. As discussed in Q5, the system needs a component that flags inputs that are *not* skin lesions (acne, scars, wrong-region photos).

4. Clinical UI/UX. My Streamlit prototype is a demo. Production needs:

- PACS / EHR integration.
- An audit log of every prediction for traceability when errors occur.
- A clear workflow defining who has final responsibility (clinician).

5. Post-deployment monitoring. Data drift detection (does the input distribution change after six months?), performance monitoring (does accuracy degrade?), a feedback loop for clinicians to correct mistakes.

6. Liability and insurance. When the model errs and harms a patient, who is responsible? The clinician? The hospital? The vendor? Contracts must clarify this before deployment.

I do not claim this thesis is production-ready. It is a *research prototype* demonstrating technical feasibility. Real deployment requires ~2 years of engineering plus clinical trials.

Q11. Defense kit checklist

A list of items to prepare before defense day.

Hardware

- Primary laptop (battery full + charger).
- USB-C / HDMI adapter for the projector.
- External mouse or presentation clicker.
- Backup USB containing: all slides + thesis PDF + study guide PDF + demo images.
- Small headphones (in case you want to replay a rehearsal recording).

Software & demo

- Streamlit demo: run `streamlit run app.py` 15 minutes before defense to warm up.
- Open the browser tab at `http://localhost:8501`.

- Set theme (Dark/Light) via menu : → Settings → Theme.
- Upload one demo image and view Grad-CAM for all 4 models to cache them.
- Have 3 demo images ready: 1 clear MEL, 1 benign NV, 1 hard case (low-confidence to demonstrate the warning).

Backup plans

- If Streamlit crashes: 5 demo screenshots (prediction + Grad-CAM) embedded in the slides.
- If the projector does not recognise the laptop: PDF slides on USB, open them on the room's machine.
- If internet fails: app.py runs 100% offline, no internet needed.

Printed materials

- Slides 4-up handout for the committee to follow.
- One printed copy of the English thesis.
- Printed study guide for your own reference (not required to hand to the committee).

One hour before defense

- Test on the actual projector (if allowed).
- Listen to the last rehearsal recording.
- Light snack (banana, almonds) — avoid hunger during the talk.
- Restroom.
- Phone to airplane mode.